1st Semester Syllabus

1. Organic structures
2. Structure of molecules
3. Organic reactions
4. Nucleophilic addition to the carbonyl group
5. Delocalization and conjugation
6. Acidity, basicity, and p*K*a
7. Using organometallic reagents to make C-C bonds
8. Nucleophilic substitution at the carbonyl (C=O) group
9. Nucleophilic substitution at C=O with loss of carbonyl oxygen
10. Stereochemistry

How to represent properly organic structures: The guidelines

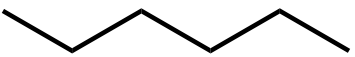
Guideline 1: Draw chains of atoms as zig-zags

Guideline 2: Miss out the H's attached to carbon atoms, along with the C-H bonds (unless there is a good reason not to)

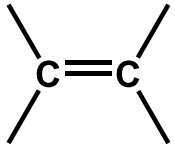
Guideline 3: Miss out the capital C's representing carbon atoms (unless there is a good reason not to)

Guideline 4: Always bear in mind that C's form 4 bonds and also that double bonds are planar (120°) and triple bonds linear (180°)

Common Functional Groups



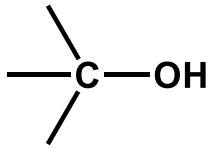
alkane



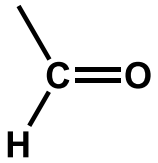
alkene



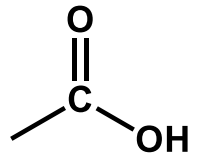
alkyne



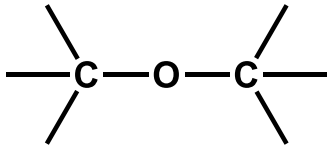
alcohol (1°, 2°, 3°)



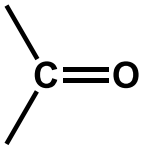
aldehyde



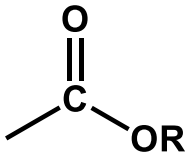
carboxylic acid



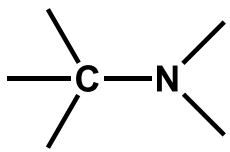
ether



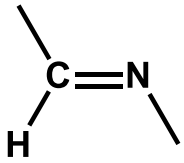
ketone



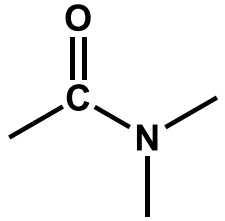
ester



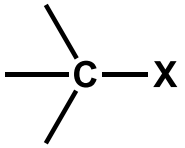
amine (1°, 2°, 3°)



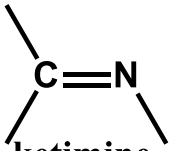
aldimine (imine)



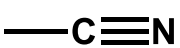
amide (1°, 2°, 3°)



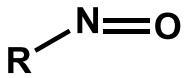
alkyl halide (1°, 2°, 3°)



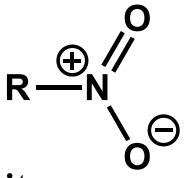
ketimine (imine)



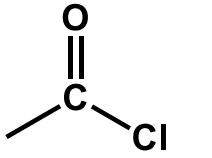
nitrile



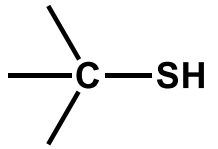
nitroso



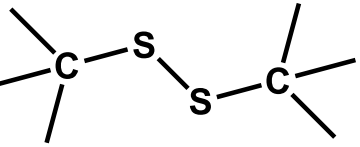
nitro compound



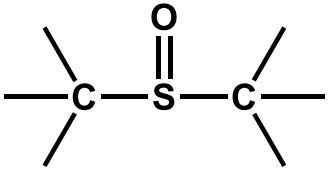
acyl chloride



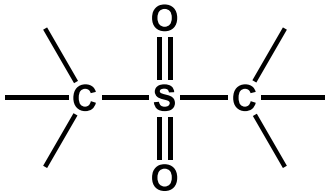
mercaptan, thiol



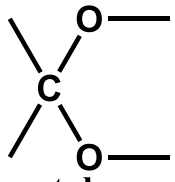
disulfide



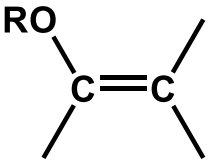
sulfoxide



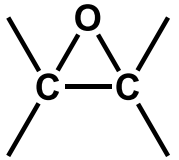
Sulfone



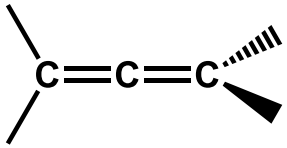
acetal



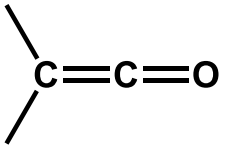
enol ether



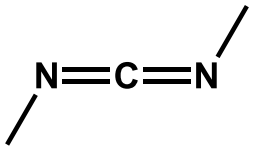
epoxide



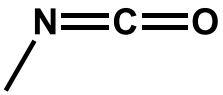
Allene



ketene



carbodiimide



isocyanate

Oxidation levels Rules

- **Bonds to hydrogen or other elements more electropositive than carbon contribute to -1 to the oxidation level.**
- **Bonds to other carbon atoms contribute 0 to the oxidation level.**
- **Bonds to oxygen or other elements more electronegative than carbon contribute +1 to the oxidation level.**
- **Multiple bounds to an element count as multiple single bonds to that element.**
- **A pair of electrons on carbon contribute -1 to the oxidation level.**
- **A positive charge on carbon contribute +1 to the oxidation level**

Naming Organic Molecules-IUPAC short summary

- **Substitutive nomenclature**

IUPAC nomenclature is based on naming a molecule's longest chain of carbons connected by single bonds, whether in a continuous chain or in a ring.

Prefixes- name of parent hydride- suffixes

All deviations, either multiple bonds or atoms other than carbon and hydrogen, are indicated by prefixes or suffixes according to a specific set of priorities.

Naming Organic Molecules-IUPAC short summary

Nomenclature of alkanes and cycloalkanes (see Lecture 1)

- **Selection of the principal chain (i.e. longest): parent hydride**

*n**	Name	*n**	Name	*n**	Name	*n**	Name
1	Methane	11	Undecane	21	henicosane	60	Hexacontane
2	Ethane	12	dodecane	22	Docosane	70	Heptacontane
3	Propane	13	tridecane	23	Tricosane	80	Octacontane
4	Butane	14	tetradecane	30	Triacontane	90	Nonacontane
5	Pentane	15	pentadecane	31	Hentriacontane	100	Hectane
6	Hexane	16	hexadecane	34	Tetratriacontane	110	Decahectane
7	Heptane	17	Heptadecane	35	Pentatricontane	120	Icosahectane
8	Octane	18	Octadecane	40	tetracontane	121	Henicosahectane
9	Nonane	19	Nonadecane	46	hexatetracontane		
10	Decane	20	Icosane	50	Pentacontane		

Naming Organic Molecules-IUPAC short summary

Nomenclature of alkanes and cycloalkanes (see Lecture 1)

- **Selection of the principal chain (i.e. longest): parent hydride**
- **Identification and naming of the side chains**
- **Determination of the position of side-chains (for alkanes prefixes) and selection of locants using the rule of lowest locants.**

Naming Organic Molecules-IUPAC short summary

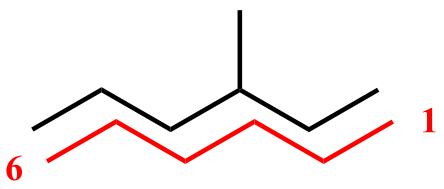
Nomenclature of alkanes and cycloalkanes (see Lecture 1)

- **Numerical or multiplicative prefixes**

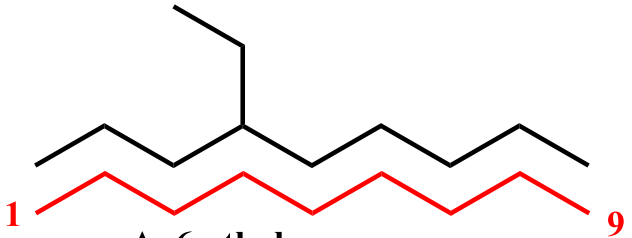
*n**	Numerical prefix	*n**	Name	*n**	Name
1	mono	10	Deca (or decakis)	19	nonadeca
2	Di (or bis)	11	Undeca	20	Docosane
3	Tri (or tris)	12	Dodeca	21	henicosa
4	Tetra (or tetrakis)	13	Trideca	23	docosa
5	Penta (or pentakis)	14	Tetradeca	30	Hentriaconta
6	Hexa (or hexakis)	15	Pentadeca	34	Tetratriaconta
7	Hepta (or heptakis)	16	Hexadeca	35	Pentatriconta
8	Octa (or ocatkis)	17	Heptadeca	40	Tetraconta
9	Nona (or nonakis)	18	octadeca	46	hexatetraconta

Naming Alkanes

- Look at the longest continuous chain, count the C's
- Find the name of the branch
- Compose the name of the molecule, the number indicating the branch should be the lowest possible
- What is the name of these molecules?



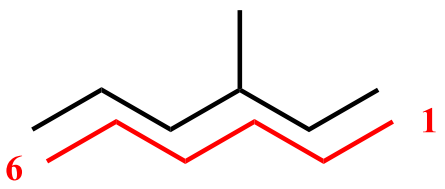
A. 2-ethylpentane
B. 4-methyhexane
C. 3-methylhexane



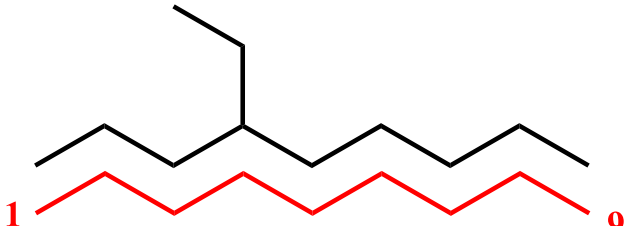
A. 6-ethylnonane
B. 3-propyloctane
C. 4-ethylnonane

Names for Branches-Simple Examples

- Look at the longest continuous chain, count the C's
- Find the name of the branch
- Compose the name of the molecule, the number indicating the branch should be the lowest possible
- What is the name of these molecules?



A. 2-ethylpentane
B. 4-methyhexane
→ C. 3-methylhexane



A. 6-ethylnonane
B. 3-propyloctane
→ C. 4-ethylnonane

Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

- Identification of the **principal functional group** and **substituents**
- Identification of the **longest continuous chain** containing the principal functional group
- Assign **locants** (*i.e.* numbering) to the **principal functional group** and **substituents**

Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

Principal Functional group

- The **principal functional group** is used to define the class the compound belongs to *e.g.* an alcohol, ROH
- The **principal functional group** is the **highest priority functional group**.
- The **principal functional group** is usually given the **lowest locant possible**.
- When a group is named as a substituent, the priority order does not apply.

Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

Longest chain

- The **longest continuous chain** containing the **principal functional group** defines the **root name**.
- Other groups attached to this chain are called **substituents**.
- If there are two chains of equal length, then the choice that gives the simplest **substituents** is chosen.

Naming Organic Molecules-IUPAC short summary

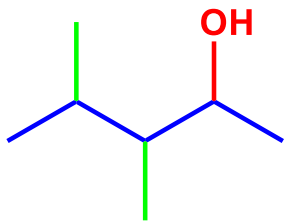
Numbering (*i.e.* assigning locants)

- The numbers that define the positions of the principal functional group and substituents are called **locants**.
- Compounds are numbered from one end of the **longest continuous chain**.
- The locants are assigned such that the **principal functional group** gets the **lowest possible locant**.
- If this results in a "tie" then the **first point of difference rule** is applied so that the first time a difference in numbering occurs, then the method that gives the lower number at this first difference is used.
- In the event that there is no first point of difference then **alphabetisation** is used.
- Note that a locant can *only* be omitted if they are a. "1" and b. would be unambiguous (see later).

Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

Locant multiplier Substituant PREFIXES ROOT Locant Functional group SUFFIXES

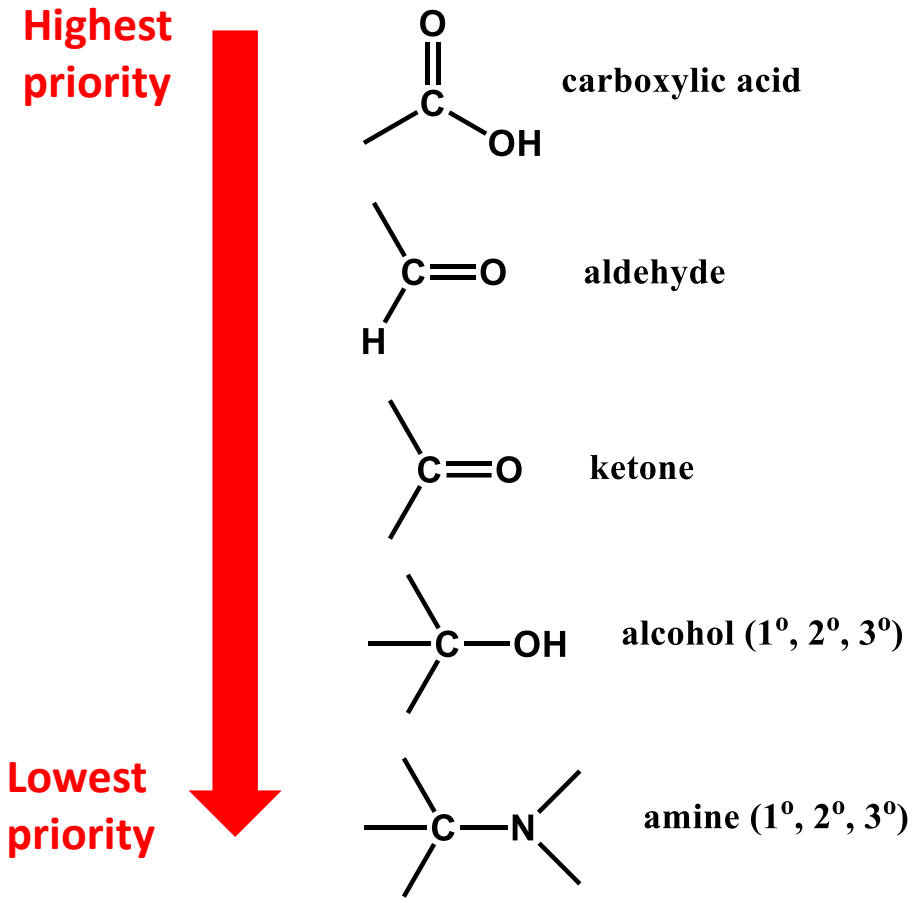


3,4-dimethyl pentan-2-ol

Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

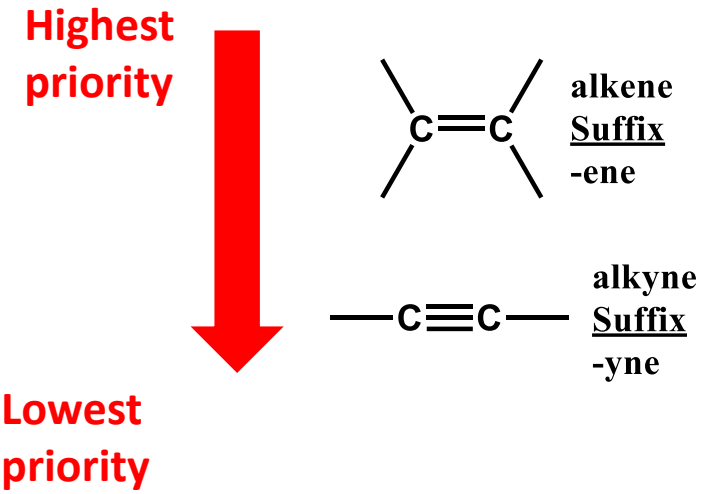
A. Priorities of substituents and functional groups indicated by prefix and suffix:



Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

B. Priorities of substituents and functional groups indicated by suffix only:



Naming Organic Molecules-IUPAC short summary

Nomenclature of molecules containing substituents and functional groups:

C. Priorities of substituents and functional groups indicated by prefix only:

equivalent
Priority
i.e. none

R—

alkane
Prefix
see review box

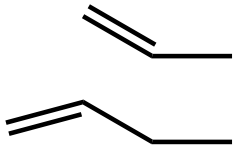
R—O—

alkoxy
Prefix
alkoxy-

Halogen
Prefix

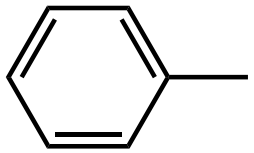
F— fluoro-
Cl— chloro-
Br— bromo-
I— iodo-

—NO_2 nitro



vinyl

allyl



phenyl

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

C. Priorities of substituents and functional groups indicated by prefix only:

- **Step 1: Find the longest continuous carbon chain and determine the root name for this parent chain**
- **Step 2: Number the chain in the direction such that the position number of the first substituent is the smaller number. If the first substituents from either end have the same number, then number so that the second substituent has the smaller number, etc.**
- **Step3: Determine the name and position number of each substituent (a substituent on a nitrogen is designated with an "*N*" instead of a number)**
- **Step 4: Indicate the number of identical groups by the prefixes di, tri, tetra, etc.**

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

C. Priorities of substituents and functional groups indicated by prefix only:

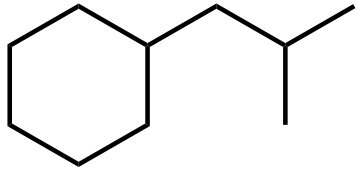
- **Step 5: Place the position numbers and names of the substituent groups in alphabetical order, before the root name. In alphabetizing, ignore prefixes like *sec*-, *tert*-, di, tri, etc., but include iso and cyclo. Always include a position number for each substituent, regardless of redundancies.**

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- **First work out the name of the complex substituent**
- The longest chain in the substituent is C3 therefore it's a propyl substituent
- There is one substituent off this chain, a C1 alkyl group *i.e.* a **methyl** group
- The methyl group is on C2 on the complex substituent

Substituent name : **2-methylpropyl**

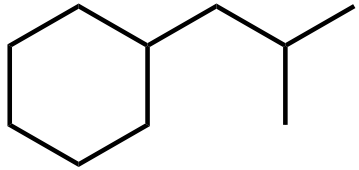


Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- **First work out the name of the complex substituent**
- The longest chain in the substituent is C3 therefore it's a propyl substituent
- There is one substituent off this chain, a C1 alkyl group *i.e.* a **methyl** group
- The methyl group is on C2 on the complex substituent

Substituent name : **2-methylpropyl**



- **Now work on the whole structure**:

Functional group is an alkane, therefore suffix = -**ane**
The ring structure is C6 therefore root = **hex**
There is one substituent: (2-methylpropyl)

- Note the syntax: no space or hypen to separate the letters of the substituent and parent root.

(2-methylpropyl)cyclohexane

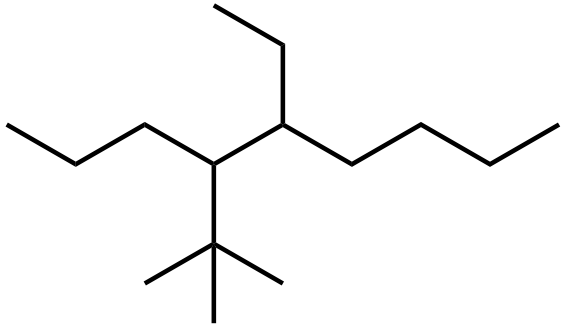
Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- **First work out the name of the complex substituent**

- The longest chain in the substituent is C2 therefore it's an ethyl substituent
- There are two substituents off this chain, both C1 alkyl group *i.e.* **methyl** groups
- Both methyl groups are on C1 on the complex substituent

- Complex substituent name : **1,1-dimethylethyl**

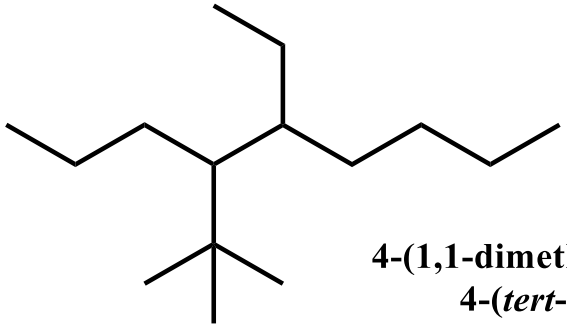


Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- Functional group is an alkane, therefore suffix = -**ane**
  The longest continuous chain is C9 therefore root = **non**
  There are two substituents:
  simple substituent is a C2 alkyl group *i.e.* an **ethyl** group
- a complex substituent (see above) = (1,1-dimethylethyl)

- The first point of difference rule requires numbering from the *left* as drawn, the locants are **4-** and **5-**
  Alphabetisation requires that the complex substituent be listed first since "d" is before "e".

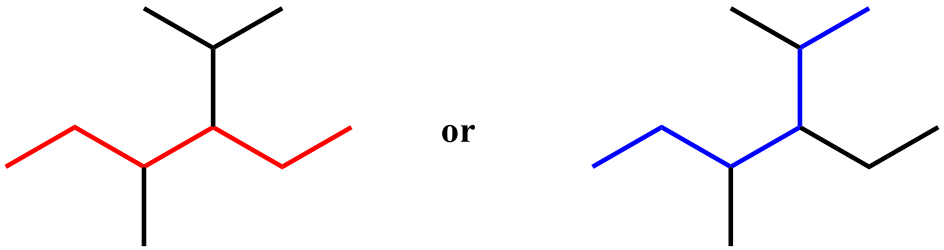


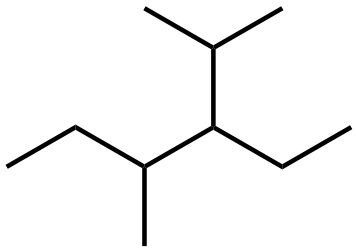
4-(1,1-dimethylethyl)-5-ethylnonane or
4-(*tert*-butyl)-5-ethylnonane

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- Remember that if a structure has two chains of equal length, then **the choice that gives the simplest branches is chosen**

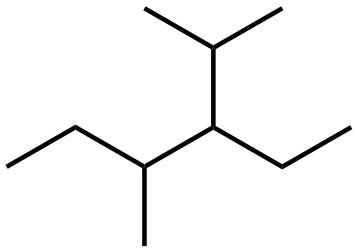




Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

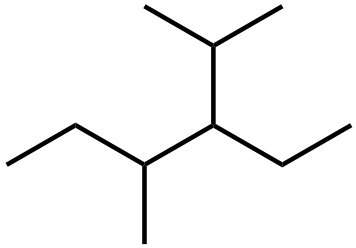
- Remember that if a structure has two chains of equal length, then **the choice that gives the simplest branches is chosen**
- Functional group is an alkane, therefore suffix = -**ane**
  The longest continuous chain is C6 therefore root = **hex** The **lower choice** is correct



Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- Remember that if a structure has two chains of equal length, then **the choice that gives the simplest branches is chosen**
- Functional group is an alkane, therefore suffix = -**ane**
  The longest continuous chain is C6 therefore root = **hex** The **lower choice** is correct
- it has more but simpler substituents There are three simple substituents a C2 alkyl group *i.e.* an **ethyl** group two C1 alkyl groups *i.e.* **methyl** groups
- The first point of difference rule requires numbering from the top *right* as drawn, the locants are **2-, 3-** and **4-**
  Alphabetisation requires that the simple substituents be listed as **ethyl** before **methyl**



3-ethyl-2,4-dimethylhexane

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

- Substituted cycloalkanes are named in an analogous fashion to regular alkanes (also referred to as acyclic alkanes, *i.e.* non-cyclic alkanes).

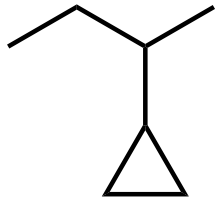
  In principle, a substituted cycloalkane could be name in two ways, either as an alkyl substituted cycloalkane, or as a cycloalkyl substituted alkane When the ring size (*e.g.* number of C atoms) is larger than the longest continuous chain, then the ring becomes the parent and hence the system is treated as a alkyl cycloalkane.

- Other rules:
  - If a ring is monosubstituted then no locant is required since the substituent must be at C1.
  - If a ring is polysubstituted then each *substituent* requires a locant to unambiguously assign their relative positions.

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

Ring is smaller than the chain, so the ***chain is the parent***.
Functional group is an alkane, therefore suffix = -**ane**



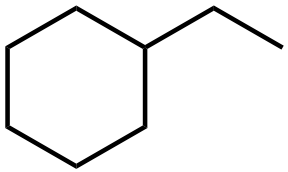
The longest continuous chain is C4 therefore root = **but**

The branch is a C3 cycloalkyl group *i.e.* a **cyclopropyl** group
The first point of difference rule requires numbering from the *right* as drawn, the substituent locant is **2-**

2-cyclopropylbutane

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents



Ring is larger than the chain, so the ***ring is the parent***.
Functional group is an alkane, therefore suffix = -**ane**

The ring is C6 therefore root = **cyclohex**

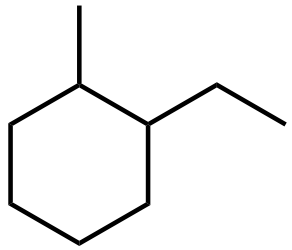
The branch is a C2 alkyl group *i.e.* a **ethyl** group
Since it's monosubstituted, no locant is required since it is unambiguous (*i.e*. the ethyl group will define the C it is attached to as C1)

ethylcyclohexane

Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

Ring is larger than the chain, so the ***ring is the parent***.
Functional group is an alkane, therefore suffix = -**ane**



The ring is C6 therefore root = **cyclohex**

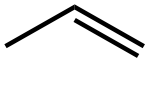
One branch is a C2 alkyl group *i.e.* an **ethyl** group
The other branch is a C1 alkyl group, *i.e.* a **methyl** group
Since the first point of difference doesn't resolve the numbering, the alphabetisation order is used.

1-ethyl-2-methylcyclohexane

Naming Organic Molecules

Alkenes

- The root name is based on the longest chain containing both ends of the alkene unit, the C=C.
- The chain is numbered so as to give the alkene unit the lowest possible numbers.
- The locant for the first carbon of the alkene is used in the name.
- If the position of the alkene is unambiguous, the locant is not required



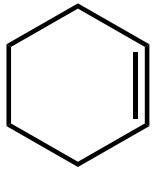
prop-1-ene



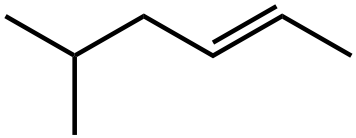
(*E*)-but-2-ene



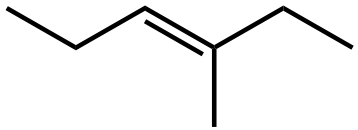
but-1-ene



cyclohexene



(*E*)-5-methylhex-2-ene

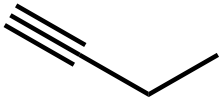


(*E*)-3-methylhex-3-ene

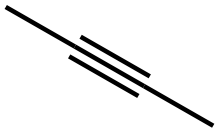
Naming Organic Molecules

Alkynes

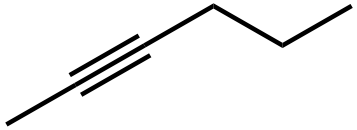
- The root name is based on the longest chain containing both ends of the alkyne unit, the C≡C.
- The chain is numbered so as to give the alkyne unit the lowest possible number.



but-1-yne



but-2-yne



hex-2-yne

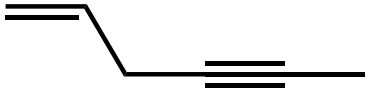
Naming Organic Molecules

Enynes

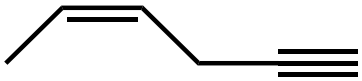
- The term enyne simply implies the presence of **both** an alkene and an alkyne.
  Enynes are named in a similar manner to alkenes, alkynes and polyenes. The root name is based on the longest chain containing **both ends** of the alkene and alkyne units.
- The alkene suffix "ene" minus the last "e" is inserted after the root with its locant before the root.
- The alkyne suffix, "yne" is added at the end with its locant preceding it. Hence alkene + alkyne = enyne.
- The chain is numbered in accord with the first point of difference rule to either the alkene or alkyne units the lowest possible locant.
- The locant for the lowest numbered carbon of the alkene or alkyne is used in the name.
- If there is a choice, then the C=C takes priority and is given the lowest locant.

Naming Organic Molecules

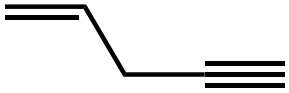
Enynes



hex-1-en-4-yne



(*Z*)-hex-4-en-1-yne



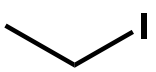
pent-1-en-4-yne

Naming Organic Molecules

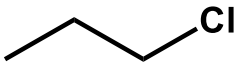
- **<u>Haloalkanes / Alkyl halides</u>**
- **Haloalkane style:**
  The root name is based on the longest chain containing the halogen.
  - This root give the alkane part of the name.
  - The type of halogen defines the halo prefix, *e.g.* **chloro**-
  - The chain is numbered so as to give the halogen the lowest possible number
- **Alkyl halide style:**
  - The root name is based on the longest chain containing the halogen.
  - This root give the alkyl part of the name.
  - The type of halogen defines the halide suffix, *e.g.* **chloride**
  - The chain is numbered so as to give the halogen the lowest possible number.

Naming Organic Molecules

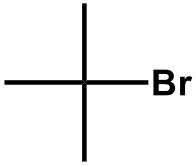
- **Haloalkanes / Alkyl halides**
- **Haloalkane style:**



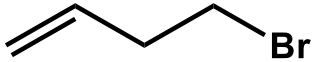
iodoethane



1-chloropropane

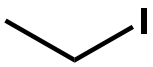


2-bromo-2-methylpropane

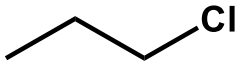


4-bromobut-1-ene

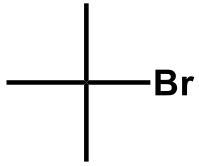
- **Alkyl halide style:**



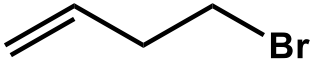
ethyl iodide



propyl chloride



tert-butyl bromide



not applicable

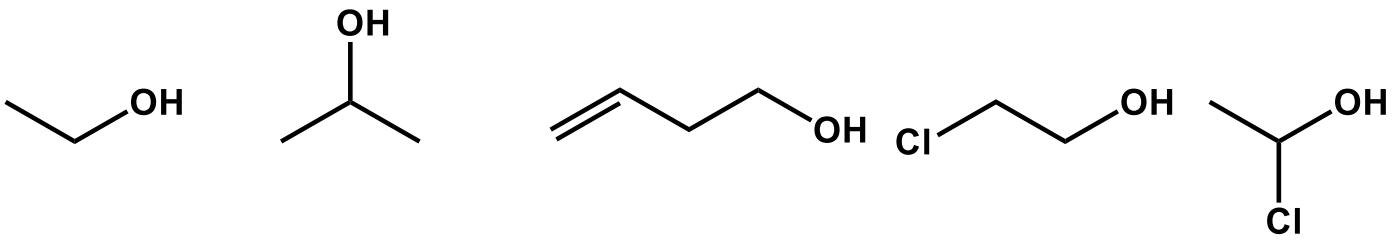
Naming Organic Molecules

- **<u>Alcohols</u>**
- Functional class name = **alkyl alcohol** *e.g.* ethyl alcohol
  Substituent suffix = -**ol** *e.g.* ethanol
  Substituent prefix = **hydroxy-** *e.g.* hydroxyethane
  Structural unit : alcohols contain R-OH

- The root name is based on the longest chain with the -OH attached. The chain is numbered so as to give the alcohol unit the lowest possible number.
  The alcohol suffix is appended after the hydrocarbon suffix minus the "e" : *e.g*. -ane + -ol = -anol or -ene + ol = -enol.

Naming Organic Molecules

- **Alcohols**



ethanol

propan-2-ol

but-3-en-1-ol

2-chloroethan-1-ol

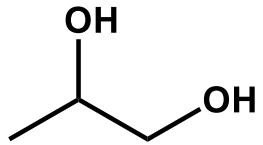
1-chloroethan-1-ol

Naming Organic Molecules

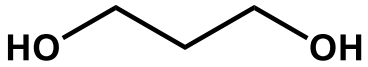
- **Diols**
- The term diol simply implies the presence of two alcohols. Polyols contain two or more -OH groups.
  The root name is based on the longest chain containing **both** the alcohol groups.
- The chain is numbered so as to give the one of the alcohol groups the lowest possible number (*i.e*. first point of difference).
- The appropriate multiplier (*i.e.* di- for two, tri for three *etc*.) is inserted before the -ol suffix or before the root.

Naming Organic Molecules

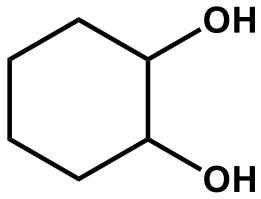
- **Diols**



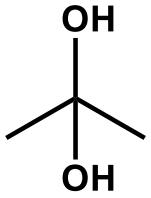
propane-1,2-diol



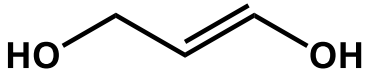
propane-1,3-diol



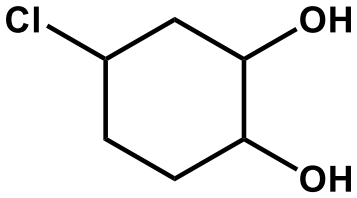
cyclohexane-1,2-diol



propane-2,2-diol



(*E*)-prop-1-ene-1,3-diol



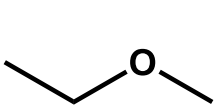
4-chlorocyclohexane-1,2-diol

Naming Organic Molecules

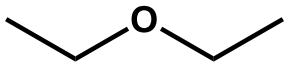
- **Ethers**
- Functional class name = alkyl alkyl ether *e.g.* ethyl methyl ether
- Substituent prefix = **alkoxy-** *e.g.* methoxyethane
- **"Simple" ethers**
  If both groups are simple alkyl groups, then the ether is usually named as **alkyl alkyl ether**
- The alkyl groups are listed in alphabetical order
- If the two alkyl groups are the same, then it's a **dialkyl ether**
- **"Intermediate" ethers**
  If one of the groups is more complex then the ether group is usually treated as an **alkoxy substituent** (*i.e.* R-O-).
- The ether is treated as a substituent (*i.e.* it is not the principal functional group).
- The more complex group (*i.e.* longer chain, principal functional group, more branched, other substituents) defines the root.

Naming Organic Molecules

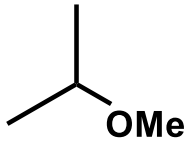
- **Ethers**
- Functional class name = alkyl alkyl ether *e.g.* ethyl methyl ether
- Substituent prefix = **alkoxy-** *e.g.* methoxyethane



methoxyethane
or ethyl methyl ether



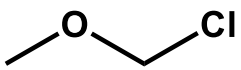
ethoxyethane
or diethyl ether



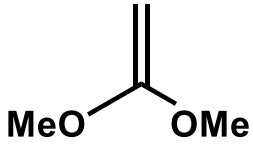
2-methoxypropane



4-methoxybut-1-ene



chloro(methoxy)methane



1,1-dimethoxyethene

Naming Organic Molecules

- **<u>Amines</u>**
- Functional class name = **alkylamines** or **alkanamines**

  Substituent suffix = -**amine** *e.g.* ethylamine

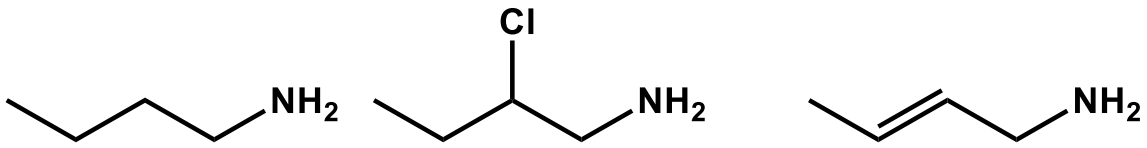
  Substituent prefix = **amino-** *e.g.* aminoethane

  Structural unit (depending on R, R' and R")
- primary amines contain R-NH_2
- secondary amines contain R_2NH
- tertiary amines contain R_3N.

Naming Organic Molecules

- **Amines**
- **Primary amines**
  Primary amines have *one* alkyl group attached to the N.
- The root name is based on the longest chain with the -NH_2 attached.
- The chain is numbered so as to give the amine unit the lowest possible number.

- The amine suffix is appended to the appropriate **alkyl** root or **alkana**- root.

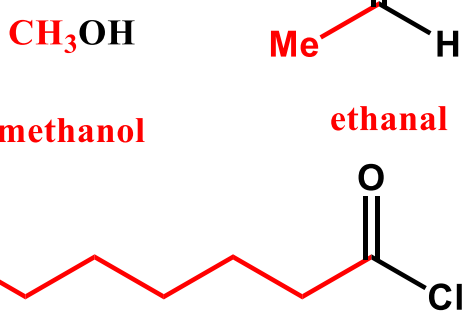


butan-1-amine **2-chlorobutan-1-amine** **(*E*)-but-2-en-1-amine**

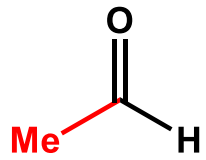
Naming Organic Molecules

- **Amines**
- **Secondary amines**
  Secondary amines have *two* alkyl groups attached to the N.
- The root name is based on the longest chain with the -NH attached.
- The chain is numbered so as to give the amine unit the lowest possible number.
- The other alkyl group is treated as a substituent, with *N* as the locant.
- The N locant is listed *before* numerical locants, *e.g*. *N*,2-dimethyl....
- The amine suffix is appended to the appropriate **alkyl** root or **alkana**- root.

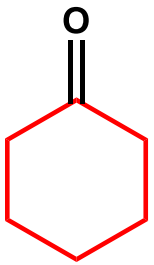
Functional Groups-Naming Organic Molecules



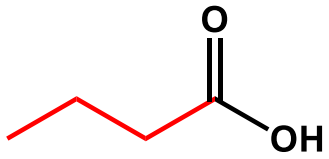
methanol



ethanal



cyclohexanone



butanoic acid

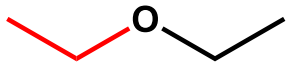


pentanenitrile

heptanoyl chloride



ethyne



ethoxyethane



propene



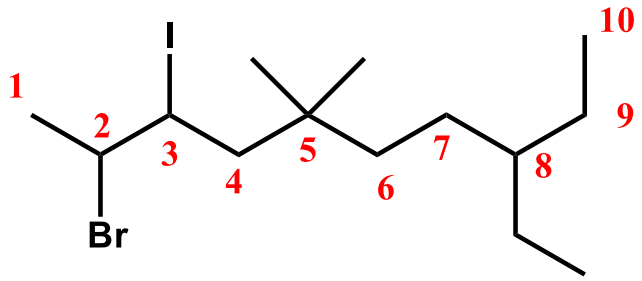
nitromethane

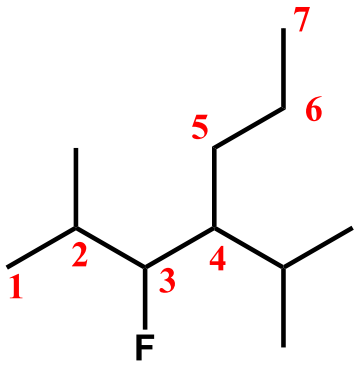
Naming Organic Molecules

Naming substituted Alkanes and Cycloalkanes- C substituents

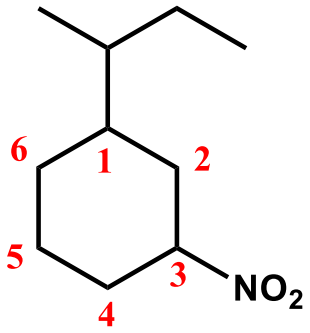
Examples:

2-bromo-8-ethyl-3-iodo-5,5-dimethyldecane





3-fluoro-4-isopropyl-2-methylheptane

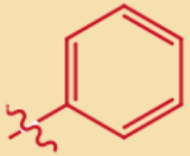
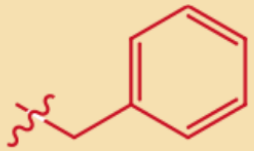
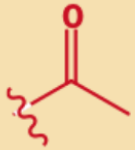
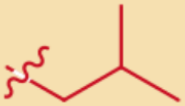
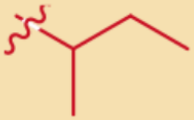


1-(*sec*-butyl)-3-nitrocyclohexane

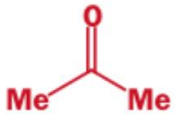
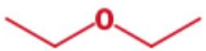
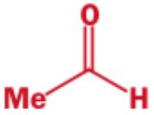
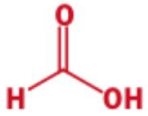
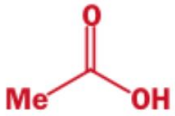
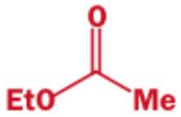
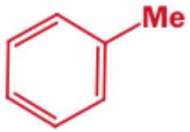
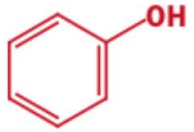
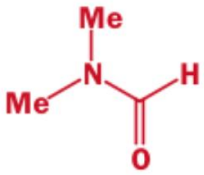
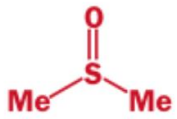
Organic Structures-Structures and Abbreviations to learn

- Names for Chains, Rings, Branches: What is the skeleton (hydrocarbon framework) of an organic molecule
- Functional Groups
- **Abbreviations used by all organic chemists: "Organic elements"**
- **Small structures to learn**

Review box: Table of fragment names and organic elements

R	alkyl		**t-Bu**	*tert*-butyl	
Me	methyl		**Ar**	aryl	any aromatic ring
Et	ethyl		**Ph**	phenyl	
Pr (or *n*-Pr)	propyl		**Bn**	benzyl	
Bu (or *n*-Bu)	butyl		**Ac**	acetyl	
***i*-Pr**	isopropyl			vinyl	
***i*-Bu**	isobutyl			allyl	
***s*-Bu**	*sec*-butyl		**X**	halide	**F, Cl, Br, or I**

Important structures to learn

acetone	
ether or diethyl ether	
acetaldehyde	
formic acid	
acetic acid or AcOH	
benzene	
ethyl acetate or EtOAc	
toluene	
pyridine	
phenol	
THF (tetrahydrofuran)	
DMF or (dimethylformamide) Me_2NCHO	
DMSO (dimethylsulfoxide)	

More details: http://www.chem.qmul.ac.uk/iupac/

Summary

- **Why we use particular diagrams?**
- **How organic chemists name molecules in writing and in speech?**
- **What is the skeleton of an organic molecule?**
- **What is a functional group?**
- **Some abbreviations used by all organic chemists**
- **Drawing organic molecules realistically in an easily understood style**